COMSM1302 Test 1 Reference Sheet

Power of 2	Value
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096
2^{13}	8192
2^{14}	16384
2^{15}	32768

Power of 2	Value
2^{-1}	1/2
2^{-2}	1/4
2^{-3}	1/8
2^{-4}	1/16
2^{-5}	1/32
2^{-6}	1/64
2^{-7}	1/128
2^{-8}	1/256
2^{-9}	1/512
2^{-10}	1/1024
2^{-11}	1/2048
2^{-12}	1/4096
2^{-13}	1/8192
2^{-14}	1/16384
2^{-15}	1/32768
2^{-16}	1/65536

Table 1: Reference for calculated powers of 2

zx	nx	zy	ny	f	no	out
1	0	1	0	1	0	0
1	1	1	1	1	1	1
1	1	1	0	1	0	-1
0	0	1	1	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	1	!x
1	1	0	0	0	1	!y
0	0	1	1	1	1	-x
1	1	0	0	1	1	-y
0	1	1	1	1	1	x+1
1	1	0	1	1	1	y+1
0	0	1	1	1	0	x-1
1	1	0	0	1	0	y-1
0	0	0	0	1	0	x+y
0	1	0	0	1	1	x-y
0	0	0	1	1	1	y-x
0	0	0	0	0	0	x&y
0	1	0	1	0	1	x|y

Table 2: Behaviour reference for the Hack ALU (taken from Nisan and Schocken)

Base 10	Symbol	Name
10^{15}	P	Peta-
10^{12}	T	Tera-
10^9	G	Giga-
10^6	M	Mega-
10^3	k	Kilo-
10^0	N/A	N/A
10^{-3}	m	Milli-
10^{-6}	μ	Micro-
10^{-9}	n	Nano-
10^{-12}	p	Pico-

Base 2	Symbol	Name
2^{50}	P or Pi	Peta- or pebi-
2^{40}	T or Ti	Tera- or tebi-
2^{30}	G or Gi	Giga- or gibi-
2^{20}	M or Mi	Mega- or mebi-
2^{10}	K or Ki	Kilo- or kibi-
2^0	N/A	N/A

Table 3: Reference for SI units